FalconAero

Intelligent airport disruption recovery

AI decision support that re-plans runway slots and stands during fog and disruptions, with a human approving every plan.



Team Falconix
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UAE Hackathon 2026 · Youth Track: Pioneers · Challenge 3 (GCAA) · Responsible AI & Smart Cognitive Government

THE PROBLEM

One disruption, four resources, one chain reaction

Runway slots
Spacing grows in low visibility

↓

Stands
Wrong size or still occupied

↓

Taxiing
Routes close, times stretch

↓

Departures
Late push-back blocks the gate

Fix one resource at a time and you create the next clash

Example: a delayed A380 still occupies one of only a few A380-capable stands, so the next A380 has nowhere to park and the stand plan, taxi plan and departure bank all slip.

It happens in the UAE

Apr 2024 floods

Heaviest UAE rainfall in 75 years; DXB restricted arrivals

Jun 2025 airspace closures

Gulf airspace closures diverted Dubai-bound flights

Every winter

Recurring fog and shamal sandstorms force low-visibility operations

THE SOLUTION

FalconAero: from first warning to approved recovery plan

1 — Assess

LVP and ILS category, crosswind, runway modes and capacity from live weather

2 — Validate

Re-checks the approved plan on every new event and flags conflicts

3 — Forecast

Demand vs capacity for the next 6 hours: warns before problems arrive

4 — Re-plan

Runway slots and stands solved together, emergencies first

5 — Explain

Every change in plain language, with up to 3 options side by side

6 — Notify

Alerts the operations centre duty manager

7 — Approve

Nothing is applied without a human decision

Unified recovery engine: every change is checked against all resources in one model before a human sees it.

THE PRODUCT

One screen for the airport operations centre

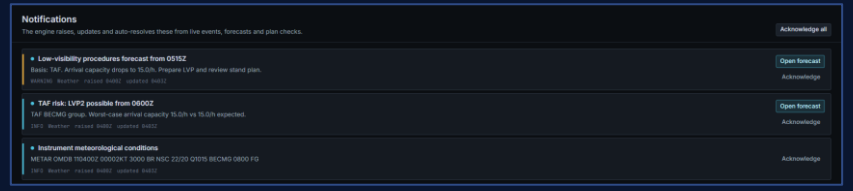


Left: flight strips, schematic airspace radar, stand timeline and alerts. Right: airport surface (M), 6-hour forecast and notifications.





6-hour forecast: demand vs capacity, predicted delay



HOW IT WORKS

One event stream in, explainable plans out

Event sources · one ingestion API

- Flight data (AODB / A-CDM)
- Weather: METAR / TAF
- NOTAMs and operator notes
- Simulator (23 scenarios)

FalconAero engine

Constraint optimisation
Google OR-Tools CP-SAT builds the plan in two stages: emergencies are locked in first, then everything else

Rule-based forecast
Demand vs capacity in 15-min bins over 6 h (METAR, then TAF) for early warning

Groq LLM
Reads NOTAMs and operator notes, writes briefings, answers questions. Never makes assignments

To the duty manager

- Up to 3 plans vs rule-based plan
- Plain-language reasons
- Notifications
- Approve / reject (logged)

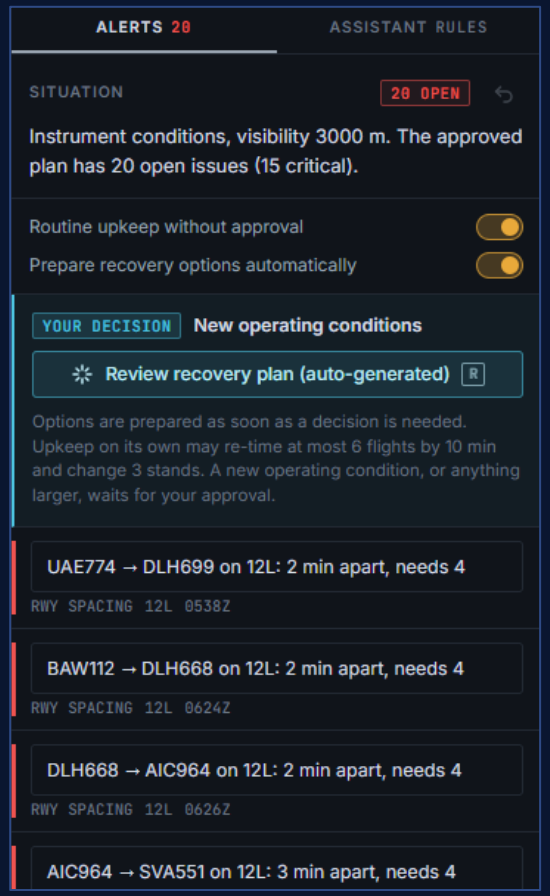
Safety rules no plan can break

- ✓ Runway usable at that time
- ✓ Spacing for the conditions
- ✓ Flow-control caps
- ✓ Airspace closures
- ✓ Aircraft fits the stand
- ✓ Stand free (10-min buffer)
- ✓ No on-block / push-back in a lightning stop
- ✓ Emergency slots locked

LIVE DEMO

Fog at Dubai International: warning to approved plan

Warning



Options



Approved



Space — Fog forms: the forecast warns before LVP begins

R — Three recovery options, computed in about 3 s

C — Full comparison against the rule-based plan

A — Duty manager approves; logged and undoable

M — Switch airspace radar / airport surface

RESULTS

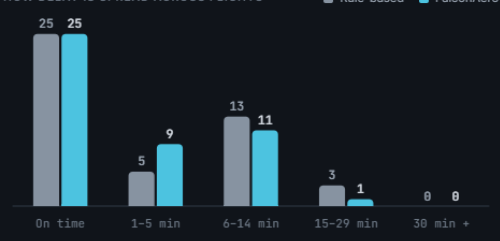
Fog scenario: fewer delays under the same rules

SAME WORLD STATE · SAME COMMITTED MOVEMENTS · SAME OPERATING RULES

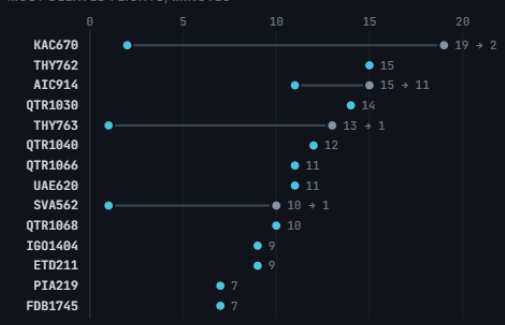
Rule-based allocation vs FalconAero (Balanced)

METRIC	APPROVED PLAN, UNRESOLVED	RULE-BASED (FIRST-COME, FIRST-SERVED)	FALCONAERO	DIFFERENCE
Critical safety issues	15	0	0	same
Flights diverted or without a slot	0	0	0	same
Average delay of served flights	1.9 min	4.2 min	3.2 min	▼ -1
Worst single delay	15 min	19 min	15 min	▼ -4
Total delay	86 min	191 min	145 min	▼ -46
Flights delayed 15 min or more	1	3	1	▼ -2
Average arrival holding	2.4 min	5.3 min	4 min	▼ -1.3
Ground waits for stands	1	0	2	▲ +2
Total taxi time	415 min	415 min	415 min	same
Arrivals on remote stands	3	3	3	same

HOW DELAY IS SPREAD ACROSS FLIGHTS



MOST DELAYED FLIGHTS, MINUTES



Both plans obey the same operating rules (runway modes, separation, ATFM caps, ramp suspension, holding limit). The rule-based policy handles one movement at a time without look-ahead; FalconAero solves runway slots and stands together, starting from the rule-based plan. Delay figures cover served flights only, so read them with the diverted / no-slot row. Figures come from the engine's current world state; they are not measurements from live airport operations.

−24%

average delay of served flights

4.2 → 3.2 min

−46 min

total delay across the bank

191 → 145 min

3 → 1

flights delayed 15 min or more

worst single delay 19 → 15 min

0

critical issues and diversions

unrevised plan had 15 critical issues

Fair test: same world state, same committed movements, same operating rules. The optimiser starts from the rule-based plan and diversions are always shown next to delay. Solved in 3.1 s and stress-tested against the TAF worst case.

Radiation fog (dawn) scenario at OMDB, simulated traffic, Balanced plan vs first-come-first-served. Measured on the current build, 11 Nov 2026.

DATA

Built on real weather, real traffic patterns and published rules

Live OMDB weather

METAR and TAF from NOAA aviationweather.gov, parsed by the engine

Live

Traffic patterns and fleet mix

OpenSky Network (ADS-B), Bayanat UAE aviation statistics, Dubai Pulse

Calibration

Operating rules

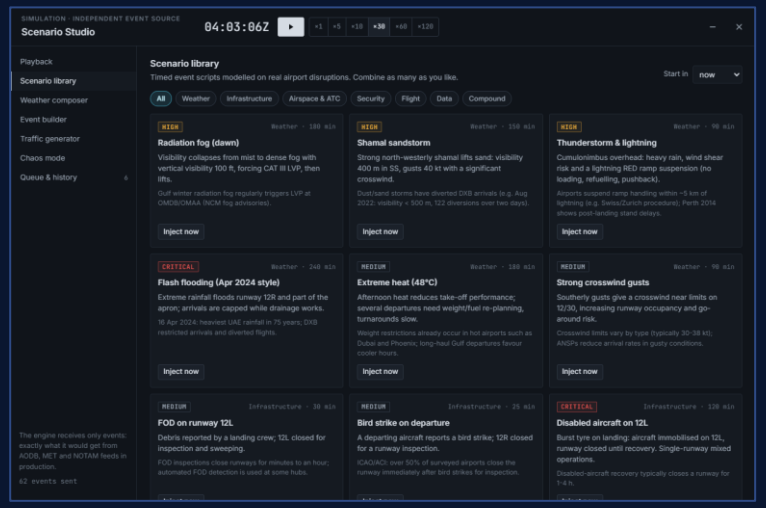
ICAO / SKYbrary guidance (LVP, ILS categories), GCAA publications

Rules

23 disruption scenarios

Modelled on real events: fog, shamal, Apr 2024 floods, Jun 2025 airspace closures, drones, emergencies

Scenarios



- Traffic in the demo is simulated and labelled
- Stand occupancy and A-CDM data are the first pilot data requests

Production-ready ingestion: every source, real or simulated, enters through the same event API, so a live AODB or A-CDM feed replaces the simulator without changing the engine.

90-DAY PILOT

From replay to shadow mode in 12 weeks

Weeks 1–4

Set up and replay

- Agree data access and security with the airport operator and the ANSP, GCAA as sponsor
- Replay 2–3 past disruption days offline

Milestone Data-sharing agreement signed; replays running

Weeks 5–9

Shadow mode

- Run alongside the operations centre on the replays and, if agreed, a read-only live feed
- Compare FalconAero proposals with what the AOC actually did

Milestone Side-by-side log of proposals vs real decisions

Weeks 10–12

Evaluate

- Controllers score the proposals
- KPI report: delay, diversions, stand changes, acceptance rate

Milestone Go / no-go for an operational trial

Owner

GCAA sponsor · airport operator · ANSP · Team Falconix

Success KPIs

Delay and diversions vs actual · acceptance rate · stand changes

Stretch goal

Live data feeds (accreditation usually takes longer than 90 days)

MARKET

A decision layer on top of the systems airports already run

FalconAero decision support

Recovery plans, forecasts, explanations, approvals

AODB | **A-CDM** | **AMAN / DMAN** | **MET / NOTAM**

Existing airport and ANSP systems: complemented, not replaced

Who uses it

Airport operations centres

GCC hubs and secondary airports

Air navigation providers

Tower and approach units

Regulators

GCAA as sponsor and evaluator

Business model

- Annual software licence per airport or ANSP unit
- Integration services for AODB / A-CDM connection
- Scenario library and replay as a training tool

Fits UAE momentum

- Dubai Airports' AI turnaround programme
- SITA and Abu Dhabi Airports total airport management partnership
- UAE National Strategy for AI 2031

RESPONSIBLE AI

Humans stay in command

Every plan approved or rejected

Both are logged, and an approval can be undone

Clearances stay with ATC

FalconAero proposes target times; ATC issues clearances

Routine upkeep within fixed limits

Small, bounded changes only, and it can be switched off

New conditions go to the controller

Fog, closures or caps always trigger a plan for human review

The LLM never assigns

It parses text and explains; the optimiser plans under hard safety rules

Honest by design

Simulated data is labelled, and limits are stated openly



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FalconAero · Team Falconix: Mohan Sri Sai Varma Alluri, Rithikh Chandran · Thank you